

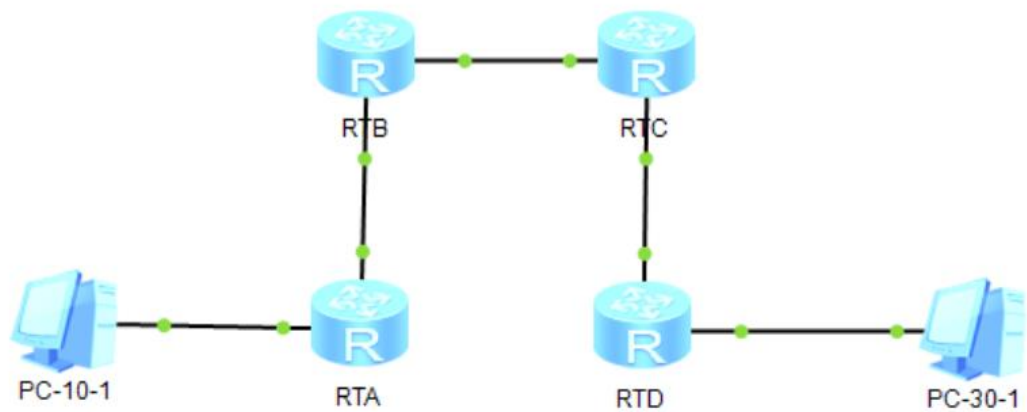
Lab-6 Basic OSPF Configuration and Analysis

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一、Lab-6-1: Basic Multiple Area OSPF Configuration

Q6-1.1~6-1.16:

(1) 创建的拓补图截图如下：



(2) RTA:

1. OSPF routing table

```
[RTA]display ospf routing

      OSPF Process 1 with Router ID 10.1.1.1
        Routing Tables

Routing for Network
Destination      Cost   Type      NextHop      AdvRouter     Area
172.16.101.0/24  1      Transit   172.16.101.1  10.1.1.1      0.0.0.1
172.16.102.0/24  2      Inter-area 172.16.101.2  10.2.2.2      0.0.0.1
172.16.103.0/24  3      Inter-area 172.16.101.2  10.2.2.2      0.0.0.1
192.168.30.0/24  4      Inter-area 172.16.101.2  10.2.2.2      0.0.0.1
192.168.200.100/32 3      Inter-area 172.16.101.2  10.2.2.2      0.0.0.1

Total Nets: 5
Intra Area: 1  Inter Area: 4  ASE: 0  NSSA: 0
```

2. Link-State Database

```
[RTA]display ospf lsdb

      OSPF Process 1 with Router ID 10.1.1.1
        Link State Database

Area: 0.0.0.1
Type      LinkState ID      AdvRouter      Age  Len  Sequence      Metric
Router    10.2.2.2           10.2.2.2       578  36   80000003      1
Router    10.1.1.1           10.1.1.1       570  36   80000005      1
Network   172.16.101.1       10.1.1.1       570  32   80000002      0
Sum-Net   192.168.200.100    10.2.2.2       400  28   80000001      2
Sum-Net   172.16.103.0       10.2.2.2       492  28   80000001      2
Sum-Net   192.168.30.0       10.2.2.2       415  28   80000001      3
Sum-Net   172.16.102.0       10.2.2.2       581  28   80000001      1
```

3. OSPF neighbor

```
[RTA]display ospf peer

      OSPF Process 1 with Router ID 10.1.1.1
        Neighbors

Area 0.0.0.1 interface 172.16.101.1(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.2.2.2      Address: 172.16.101.2
  State: Full  Mode:Nbr is Master  Priority: 1
  DR: 172.16.101.1  BDR: 172.16.101.2  MTU: 0
  Dead timer due in 40 sec
  Retrans timer interval: 5
  Neighbor is up for 00:16:26
  Authentication Sequence: [ 0 ]
```

4. updates of the LSAs that the LSDB receives

```
[RTA]display ospf statistics updated-lsa

      OSPF Process 1 with Router ID 10.1.1.1
        Statistics of Received LSAs

Begin time: 2025-05-10 14:43:11-08:00

AdvRouter      Total      Updated at
10.2.2.2        6          2025-05-10 14:46:10-08:00
```

5. route calculation statistics on the router

```
[RTA]display ospf spf-statistics

      OSPF Process 1 with Router ID 10.1.1.1
Routing table change statistics:
Date           Time           Intra    Inter    External  Reason
2025-05-10 14:46:10-08:00      0        1        0        LSA
2025-05-10 14:45:54-08:00      0        1        0        LSA
2025-05-10 14:44:38-08:00      0        1        0        LSA
2025-05-10 14:43:12-08:00      0        1        0        Topo
2025-05-10 14:43:11-08:00      0        0        0        LSA
2025-05-10 14:41:56-08:00      0        0        0        LSA
```

(3) RTB:

1. OSPF 路由表

```
[RTB]display ospf routing

      OSPF Process 1 with Router ID 10.2.2.2
        Routing Tables

Routing for Network
Destination    Cost  Type      NextHop      AdvRouter     Area
172.16.101.0/24  1    Transit  172.16.101.2  10.2.2.2      0.0.0.1
172.16.102.0/24  1    Transit  172.16.102.1  10.2.2.2      0.0.0.0
172.16.103.0/24  2    Inter-area 172.16.102.2  10.3.3.3      0.0.0.0
192.168.30.0/24  3    Inter-area 172.16.102.2  10.3.3.3      0.0.0.0
192.168.200.100/32 2    Inter-area 172.16.102.2  10.3.3.3      0.0.0.0

Total Nets: 5
Intra Area: 2  Inter Area: 3  ASE: 0  NSSA: 0
```

2. Link-State Database

```
[RTB]display ospf lsdb

      OSPF Process 1 with Router ID 10.2.2.2
      Link State Database

      Area: 0.0.0.0
  Type      LinkState ID      AdvRouter      Age  Len  Sequence      Metric
  Router    10.3.3.3          10.3.3.3       544  36   800000004     1
  Router    10.2.2.2          10.2.2.2       535  36   800000006     1
  Network   172.16.102.1      10.2.2.2       535  32   800000002     0
  Sum-Net   192.168.200.100   10.3.3.3       442  28   800000001     1
  Sum-Net   172.16.103.0      10.3.3.3       533  28   800000001     1
  Sum-Net   192.168.30.0      10.3.3.3       457  28   800000001     2
  Sum-Net   172.16.101.0      10.2.2.2       622  28   800000001     1

      Area: 0.0.0.1
  Type      LinkState ID      AdvRouter      Age  Len  Sequence      Metric
  Router    10.2.2.2          10.2.2.2       619  36   800000003     1
  Router    10.1.1.1          10.1.1.1       613  36   800000005     1
  Network   172.16.101.1      10.1.1.1       613  32   800000002     0
  Sum-Net   192.168.200.100   10.2.2.2       441  28   800000001     2
  Sum-Net   172.16.103.0      10.2.2.2       532  28   800000001     2
  Sum-Net   192.168.30.0      10.2.2.2       456  28   800000001     3
  Sum-Net   172.16.102.0      10.2.2.2       622  28   800000001     1
```

3. OSPF neighbor

```
[RTB]display ospf peer

      OSPF Process 1 with Router ID 10.2.2.2
      Neighbors

Area 0.0.0.0 interface 172.16.102.1(GigabitEthernet0/0/1)'s neighbors
Router ID: 10.3.3.3      Address: 172.16.102.2
  State: Full Mode:Nbr is Master Priority: 1
  DR: 172.16.102.1 BDR: 172.16.102.2 MTU: 0
  Dead timer due in 40 sec
  Retrans timer interval: 5
  Neighbor is up for 00:19:05
  Authentication Sequence: [ 0 ]

      Neighbors

Area 0.0.0.1 interface 172.16.101.2(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.1.1.1      Address: 172.16.101.1
  State: Full Mode:Nbr is Slave Priority: 1
  DR: 172.16.101.1 BDR: 172.16.101.2 MTU: 0
  Dead timer due in 31 sec
  Retrans timer interval: 5
  Neighbor is up for 00:20:21
  Authentication Sequence: [ 0 ]
```

4. updates of the LSAs that the LSDB receives

```
[RTB]display ospf statistics updated-lsa history

      OSPF Process 1 with Router ID 10.2.2.2
      History Information for Received LSAs
```

5. route calculation statistics on the router

```
[RTB]display ospf spf-statistics

      OSPF Process 1 with Router ID 10.2.2.2
Routing table change statistics:
Date           Time           Intra   Inter   External Reason
2025-05-10 14:46:09-08:00      0       1       0      LSA
2025-05-10 14:45:54-08:00      0       1       0      LSA
2025-05-10 14:44:37-08:00      0       1       0      LSA
2025-05-10 14:44:27-08:00      0       0       0     Topo
2025-05-10 14:44:27-08:00      0       0       0      LSA
2025-05-10 14:43:11-08:00      0       0       0     Topo
2025-05-10 14:43:11-08:00      0       0       0      LSA
2025-05-10 14:43:08-08:00      0       0       0      LSA
2025-05-10 14:42:47-08:00      0       0       0      LSA
[RTB]
```

(4) RTC:

1. OSPF 路由表

```
[RTC]display ospf routing

      OSPF Process 1 with Router ID 10.3.3.3
      Routing Tables

Routing for Network
Destination      Cost   Type      NextHop      AdvRouter     Area
172.16.102.0/24  1      Transit   172.16.102.2  10.3.3.3      0.0.0.0
172.16.103.0/24  1      Transit   172.16.103.2  10.3.3.3      0.0.0.2
172.16.101.0/24  2      Inter-area 172.16.102.1  10.2.2.2      0.0.0.0
192.168.30.0/24  2      Stub      172.16.103.1  10.4.4.4      0.0.0.2
192.168.200.100/32 1      Stub      172.16.103.1  10.4.4.4      0.0.0.2

Total Nets: 5
Intra Area: 4 Inter Area: 1 ASE: 0 NSSA: 0
```

2. Link-State Database

```
[RTC]display ospf lsd

      OSPF Process 1 with Router ID 10.3.3.3
      Link State Database

Area: 0.0.0.0
Type      LinkState ID  AdvRouter      Age  Len  Sequence      Metric
Router    10.3.3.3      10.3.3.3        1292 36   800000004      1
Router    10.2.2.2      10.2.2.2        1285 36   800000006      1
Network   172.16.102.1  10.2.2.2        1285 32   800000002      0
Sum-Net   192.168.200.100 10.3.3.3        1190 28   800000001      1
Sum-Net   172.16.103.0  10.3.3.3        1281 28   800000001      1
Sum-Net   192.168.30.0  10.3.3.3        1205 28   800000001      2
Sum-Net   172.16.101.0  10.2.2.2        1372 28   800000001      1

Area: 0.0.0.2
Type      LinkState ID  AdvRouter      Age  Len  Sequence      Metric
Router    10.3.3.3      10.3.3.3        1212 36   800000005      1
Router    10.4.4.4      10.4.4.4        1163 60   800000006      1
Network   172.16.103.2  10.3.3.3        1212 32   800000002      0
Sum-Net   172.16.102.0  10.3.3.3        1281 28   800000001      1
Sum-Net   172.16.101.0  10.3.3.3        1281 28   800000001      2
```

3. OSPF neighbor

```
[RTC]display ospf peer

      OSPF Process 1 with Router ID 10.3.3.3
        Neighbors

Area 0.0.0.0 interface 172.16.102.2(GigabitEthernet0/0/1)'s neighbors
Router ID: 10.2.2.2      Address: 172.16.102.1
  State: Full Mode:Nbr is Slave Priority: 1
  DR: 172.16.102.1 BDR: 172.16.102.2 MTU: 0
  Dead timer due in 40 sec
  Retrans timer interval: 5
  Neighbor is up for 00:21:55
  Authentication Sequence: [ 0 ]

      Neighbors

Area 0.0.0.2 interface 172.16.103.2(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.4.4.4      Address: 172.16.103.1
  State: Full Mode:Nbr is Master Priority: 1
  DR: 172.16.103.2 BDR: 172.16.103.1 MTU: 0
  Dead timer due in 38 sec
  Retrans timer interval: 5
  Neighbor is up for 00:20:39
  Authentication Sequence: [ 0 ]
```

4. updates of the LSAs that the LSDB receives

```
[RTC]display ospf statistics updated-lsa

      OSPF Process 1 with Router ID 10.3.3.3
        Statistics of Received LSAs

Begin time: 2025-05-10 14:44:27-08:00

AdvRouter      Total      Updated at
10.2.2.2        6      2025-05-10 14:44:35-08:00
10.4.4.4        5      2025-05-10 14:46:37-08:00
```

5. route calculation statistics on the router

```
[RTC]display ospf spf-statistics

      OSPF Process 1 with Router ID 10.3.3.3
Routing table change statistics:
Date           Time           Intra    Inter    External Reason
2025-05-10 14:46:09-08:00      1        0        0      LSA
2025-05-10 14:45:54-08:00      1        0        0      LSA
2025-05-10 14:45:44-08:00      0        0        0     Topo
2025-05-10 14:45:43-08:00      0        0        0      LSA
2025-05-10 14:44:38-08:00      0        0        0      LSA
2025-05-10 14:44:38-08:00      0        0        0     Topo
2025-05-10 14:44:27-08:00      0        1        0     Topo
2025-05-10 14:44:27-08:00      0        0        0      LSA
2025-05-10 14:44:17-08:00      0        0        0      LSA
```

(5) RTD:

1. OSPF 路由表

```
[RTD]display ospf routing

      OSPF Process 1 with Router ID 10.4.4.4
      Routing Tables

Routing for Network
Destination      Cost   Type      NextHop      AdvRouter      Area
172.16.103.0/24  1      Transit   172.16.103.1  10.4.4.4        0.0.0.2
192.168.30.0/24  1      Stub      192.168.30.1  10.4.4.4        0.0.0.2
192.168.200.100/32 0      Stub      192.168.200.100 10.4.4.4        0.0.0.2
172.16.101.0/24   3      Inter-area 172.16.103.2  10.3.3.3        0.0.0.2
172.16.102.0/24   2      Inter-area 172.16.103.2  10.3.3.3        0.0.0.2

Total Nets: 5
Intra Area: 3  Inter Area: 2  ASE: 0  NSSA: 0
```

2. Link-State Database

```
[RTD]display ospf lsdb

      OSPF Process 1 with Router ID 10.4.4.4
      Link State Database

Area: 0.0.0.2
Type      LinkState ID      AdvRouter      Age   Len   Sequence      Metric
Router    10.3.3.3           10.3.3.3       1357  36    800000005      1
Router    10.4.4.4           10.4.4.4       1306  60    800000006      1
Network   172.16.103.2       10.3.3.3       1357  32    800000002      0
Sum-Net   172.16.102.0       10.3.3.3       1425  28    800000001      1
Sum-Net   172.16.101.0       10.3.3.3       1425  28    800000001      2
```

3. OSPF neighbor

```
[RTD]display ospf peer

      OSPF Process 1 with Router ID 10.4.4.4
      Neighbors

Area 0.0.0.2 interface 172.16.103.1(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.3.3.3      Address: 172.16.103.2
State: Full  Mode:Nbr is Slave  Priority: 1
DR: 172.16.103.2  BDR: 172.16.103.1  MTU: 0
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00:22:54
Authentication Sequence: [ 0 ]
```

4. updates of the LSAs that the LSDB receives

```
[RTD]display ospf statistics updated-lsa

      OSPF Process 1 with Router ID 10.4.4.4
      Statistics of Received LSAs

Begin time: 2025-05-10 14:45:43-08:00

AdvRouter      Total      Updated at
10.3.3.3        7          2025-05-10 14:45:47-08:00
```

5. route calculation statistics on the router

```
[RTD]display ospf spf-statistics
```

```
OSPF Process 1 with Router ID 10.4.4.4
```

```
Routing table change statistics:
```

Date	Time	Intra	Inter	External	Reason
2025-05-10	14:46:10-08:00	0	0	0	LSA
2025-05-10	14:45:54-08:00	0	0	0	LSA
2025-05-10	14:45:44-08:00	0	2	0	Topo
2025-05-10	14:45:43-08:00	0	0	0	LSA
2025-05-10	14:45:37-08:00	0	0	0	LSA

```
[RTD]
```

(6) PC-10-1 和 PC-30-1 可以互相 ping 通:

```
PC>ping 192.168.10.11
```

```
Ping 192.168.10.11: 32 data bytes, Press Ctrl_C to break
From 192.168.10.11: bytes=32 seq=1 ttl=124 time=31 ms
From 192.168.10.11: bytes=32 seq=2 ttl=124 time=31 ms
From 192.168.10.11: bytes=32 seq=3 ttl=124 time=16 ms
From 192.168.10.11: bytes=32 seq=4 ttl=124 time=31 ms
From 192.168.10.11: bytes=32 seq=5 ttl=124 time=31 ms
```

```
--- 192.168.10.11 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 16/28/31 ms
```

```
PC>ping 192.168.30.11
```

```
Ping 192.168.30.11: 32 data bytes, Press Ctrl_C to break
From 192.168.30.11: bytes=32 seq=1 ttl=124 time=31 ms
From 192.168.30.11: bytes=32 seq=2 ttl=124 time=15 ms
From 192.168.30.11: bytes=32 seq=3 ttl=124 time=32 ms
From 192.168.30.11: bytes=32 seq=4 ttl=124 time=31 ms
From 192.168.30.11: bytes=32 seq=5 ttl=124 time=31 ms
```

```
--- 192.168.30.11 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 15/28/32 ms
```

(7) PC-10-1 和 loopback 0 可以互相 ping 通:

```
PC>ping 192.168.200.100
```

```
Ping 192.168.200.100: 32 data bytes, Press Ctrl_C to break
From 192.168.200.100: bytes=32 seq=1 ttl=252 time=31 ms
From 192.168.200.100: bytes=32 seq=2 ttl=252 time=47 ms
From 192.168.200.100: bytes=32 seq=3 ttl=252 time=16 ms
From 192.168.200.100: bytes=32 seq=4 ttl=252 time=31 ms
From 192.168.200.100: bytes=32 seq=5 ttl=252 time=31 ms
```

```
--- 192.168.200.100 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 16/31/47 ms
```



```
PC>tracert 192.168.30.11

tracert to 192.168.30.11, 8 hops max
(ICMP), press Ctrl+C to stop
 1  192.168.10.1      15 ms  16 ms  16 ms
 2  172.16.101.2     15 ms  31 ms  16 ms
 3  172.16.102.2     31 ms  16 ms  16 ms
 4  172.16.103.1     31 ms  31 ms  31 ms
 5  192.168.30.11    16 ms  31 ms  31 ms
```

1. Hello 报文，用于发现邻居
2. DD 报文，用于交换 LSDB 的摘要信息
3. LSR 报文，用于请求对方发送缺失的 LSA
4. LSU 报文，携带完整的 LSA 信息，用于填充对方的 LSDB
5. LSAck 报文，用于确认收到 LSU 报文，确保可靠性

no.	time	Source	Destination	Protocol	Length	Info
16	35.782000	172.16.101.2	224.0.0.5	OSPF	82	Hello Packet
17	35.797000	172.16.101.1	172.16.101.2	OSPF	86	DB Description
18	35.797000	172.16.101.1	224.0.0.5	OSPF	98	LS Update
19	35.813000	172.16.101.2	172.16.101.1	OSPF	70	LS Request
20	35.829000	172.16.101.2	172.16.101.1	OSPF	106	DB Description
21	35.829000	172.16.101.1	172.16.101.2	OSPF	98	LS Update
22	35.829000	172.16.101.2	224.0.0.5	OSPF	78	LS Acknowledge
23	35.829000	172.16.101.1	172.16.101.2	OSPF	82	LS Request
24	35.829000	172.16.101.1	172.16.101.2	OSPF	66	DB Description
25	35.844000	172.16.101.2	172.16.101.1	OSPF	126	LS Update
26	35.844000	172.16.101.2	224.0.0.5	OSPF	130	LS Update
27	36.579000	172.16.101.1	224.0.0.5	OSPF	118	LS Acknowledge
28	36.797000	172.16.101.1	224.0.0.5	OSPF	98	LS Update
29	36.829000	172.16.101.2	172.16.101.1	OSPF	78	LS Acknowledge

```

> Frame 20: 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on 0
> Ethernet II, Src: HuaweiTechno_ce:08:78 (00:e0:fc:00:00:00), Dst: 01:00:5c:00:12:00, Type: OSPF (89)
> Internet Protocol Version 4, Src: 172.16.101.2, Dst: 224.0.0.5
> Open Shortest Path First

```

1. 触发 LSU 报文，通过 LSU 报文将更新后的 LSA 泛洪发送到整个区域
2. LSA 老化与更新，其他路由器收到更新后，重新运行 SPF 算法计算最短路径属

61	82.453000	172.16.102.2	224.0.0.5	OSPF	82	Hello Packet
62	83.843000	172.16.102.1	255.255.255.255	RIPv1	86	Response
63	87.546000	172.16.102.2	224.0.0.5	OSPF	98	LS Update
64	87.656000	172.16.102.1	224.0.0.5	OSPF	78	LS Acknowledge
65	88.234000	172.16.102.1	224.0.0.5	OSPF	82	Hello Packet
66	91.609000	172.16.102.2	224.0.0.5	OSPF	82	Hello Packet
67	95.296000	172.16.102.1	224.0.0.5	OSPF	130	LS Update
68	95.812000	172.16.102.2	224.0.0.5	OSPF	98	LS Acknowledge
69	98.546000	172.16.102.2	224.0.0.5	OSPF	90	LS Update
70	98.671000	172.16.102.1	224.0.0.5	OSPF	78	LS Acknowledge
71	98.828000	172.16.102.1	224.0.0.5	OSPF	82	Hello Packet
72	100.781000	172.16.102.2	224.0.0.5	OSPF	82	Hello Packet

```

> Frame 68: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface 0
> Ethernet II, Src: HuaweiTechno_af:23:40 (00:e0:fc:af:23:40), Dst: IPv4mcast_05
> Internet Protocol Version 4, Src: 172.16.102.2, Dst: 224.0.0.5
√ Open Shortest Path First
  > OSPF Header
  > LSA-type 2 (Network-LSA), len 32
  > LSA-type 1 (Router-LSA), len 36

```


(11) RTB:

GE0/0/0 发送/接收 Hello、DD、LSR、LSU、LSAck 报文。目标地址是 224.0.0.5 (所有的 OSPF 路由器)，截图如下：

193	521.734000	172.16.101.2	224.0.0.5	OSPF	82 Hello Packet
194	521.843000	172.16.101.2	224.0.0.5	OSPF	90 LS Update
195	522.250000	172.16.101.1	224.0.0.5	OSPF	78 LS Acknowledge
196	525.484000	172.16.101.2	255.255.255.255	RIPv1	86 Response
197	525.859000	172.16.101.2	224.0.0.5	OSPF	98 LS Update
198	526.250000	172.16.101.1	224.0.0.5	OSPF	78 LS Acknowledge
199	527.812000	172.16.101.1	224.0.0.5	OSPF	82 Hello Packet
200	529.265000	HuaweiTechno_5e:12:...	Broadcast	ARP	60 Who has 172.16.101.2? Tell 172.
201	529.265000	HuaweiTechno_ce:08:...	HuaweiTechno_5e:12:...	ARP	60 172.16.101.2 is at 00:e0:fc:ce:
202	531.703000	172.16.101.2	224.0.0.5	OSPF	82 Hello Packet
203	532.125000	172.16.101.1	224.0.0.5	OSPF	130 LS Update
204	532.296000	172.16.101.2	224.0.0.5	OSPF	98 LS Acknowledge
205	537.078000	172.16.101.1	224.0.0.5	OSPF	82 Hello Packet

> Frame 197: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interf
> Ethernet II, Src: HuaweiTechno_ce:08:78 (00:e0:fc:ce:08:78), Dst: IPv4mcast_05
> Internet Protocol Version 4, Src: 172.16.101.2, Dst: 224.0.0.5
> Open Shortest Path First

0000 01 00 5e 00 00 05 00 0
0010 00 54 03 ef 00 00 01 5
0020 00 05 02 04 00 40 0a 6
0030 00 00 00 00 00 00 00 6
0040 02 01 0a 02 02 02 0a 6
0050 00 24 01 00 00 01 ac 1
0060 00 01

GE0/0/1 发送/接收 Hello、DD、LSR、LSU、LSAck 报文。还可能发送 Type 3 Summary-LSA，截图如下：

171	491.344000	192.168.30.11	192.168.10.11	ICMP	106 Echo (ping) reply id=0x8303, seq=3/768, ttl=126 (request in
172	496.813000	172.16.102.1	255.255.255.255	RIPv1	86 Response
173	497.750000	172.16.102.1	224.0.0.5	OSPF	82 Hello Packet
174	498.500000	172.16.102.2	224.0.0.5	OSPF	82 Hello Packet
175	502.188000	172.16.102.1	224.0.0.5	OSPF	90 LS Update
176	502.844000	172.16.102.2	224.0.0.5	OSPF	78 LS Acknowledge
177	507.657000	172.16.102.2	224.0.0.5	OSPF	82 Hello Packet
178	508.329000	172.16.102.1	224.0.0.5	OSPF	
179	516.813000	172.16.102.2	224.0.0.5	OSPF	
180	518.938000	172.16.102.1	224.0.0.5	OSPF	
181	525.969000	172.16.102.2	224.0.0.5	OSPF	
182	529.516000	172.16.102.1	224.0.0.5	OSPF	
183	530.829000	172.16.102.1	255.255.255.255	RIPv1	

> Frame 175: 90 bytes on wire (720 bits), 90 bytes captured (720 bits)
> Ethernet II, Src: HuaweiTechno_ce:08:79 (00:e0:fc:ce:08:79), Dst: I
> Internet Protocol Version 4, Src: 172.16.102.1, Dst: 224.0.0.5
> Open Shortest Path First
 > OSPF Header
 > LS Update Packet
 Number of LSAs: 1
 > LSA-type 3 (Summary-LSA (IP network)), len 28

(12) RTC:

GE0/0/0 发送/接收 Hello、DD、LSR、LSU、LSAck 报文。还可能发送 Type 3 Summary-LSA，截图如下：

175	555.825000	172.16.103.2	224.0.0.5	OSPF	82 Hello Packet
174	557.579000	172.16.103.1	224.0.0.5	OSPF	82 Hello Packet
175	558.375000	172.16.103.1	255.255.255.255	RIPv1	106 Response
176	563.157000	172.16.103.2	224.0.0.5	OSPF	82 Hello Packet
177	568.110000	172.16.103.1	224.0.0.5	OSPF	82 Hello Packet
178	572.469000	172.16.103.2	224.0.0.5	OSPF	82 Hello Packet
179	574.766000	172.16.103.2	224.0.0.5	OSPF	118 LS Update
180	575.188000	172.16.103.1	224.0.0.5	OSPF	98 LS Acknowledge
181	578.625000	172.16.103.1	224.0.0.5	OSPF	82 Hello Packet
182	581.782000	172.16.103.2	224.0.0.5	OSPF	
183	586.375000	172.16.103.1	255.255.255.255	RIPv1	
184	589.141000	172.16.103.1	224.0.0.5	OSPF	
185	591.110000	172.16.103.2	224.0.0.5	OSPF	
186	599.672000	172.16.103.1	224.0.0.5	OSPF	

> Frame 180: 98 bytes on wire (784 bits), 98 bytes captured (784 bit
> Ethernet II, Src: HuaweiTechno_0d:02:53 (00:e0:fc:0d:02:53), Dst: :
> Internet Protocol Version 4, Src: 172.16.103.1, Dst: 224.0.0.5
> Open Shortest Path First
 > OSPF Header
 > LSA-type 3 (Summary-LSA (IP network)), len 28
 > LSA-type 3 (Summary-LSA (IP network)), len 28

GE0/0/1 发送/接收 Hello、DD、LSR、LSU、LSAck 报文。目标地址是 224.0.0.0，截图如下：

209 608.297000	172.16.102.1	255.255.255.255	RIPv1	86 Response
210 613.781000	172.16.102.2	224.0.0.5	OSPF	82 Hello Packet
211 617.594000	172.16.102.1	224.0.0.5	OSPF	82 Hello Packet
212 619.110000	172.16.102.2	224.0.0.5	OSPF	90 LS Update
213 620.125000	172.16.102.1	224.0.0.5	OSPF	78 LS Acknowledge
214 622.953000	172.16.102.2	224.0.0.5	OSPF	82 Hello Packet
215 628.188000	172.16.102.1	224.0.0.5	OSPF	82 Hello Packet
216 632.110000	172.16.102.2	224.0.0.5	OSPF	82 Hello Packet
217 634.110000	172.16.102.2	224.0.0.5	OSPF	90 LS Update
218 635.110000	172.16.102.1	224.0.0.5	OSPF	78 LS Acknowledge
219 638.766000	172.16.102.1	224.0.0.5	OSPF	82 Hello Packet
220 640.328000	172.16.102.1	255.255.255.255	RIPv1	86 Response
221 641.266000	172.16.102.2	224.0.0.5	OSPF	82 Hello Packet

> Frame 213: 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interf	0000	01 00 5e 00 00 05 00 e0
> Ethernet II, Src: HuaweiTechno_ce:08:79 (00:e0:fc:ce:08:79), Dst: IPv4mcast_05	0010	00 40 04 20 00 00 01 59
> Internet Protocol Version 4, Src: 172.16.102.1, Dst: 224.0.0.5	0020	00 05 02 05 00 2c 0a 02
> Open Shortest Path First	0030	00 00 00 00 00 00 00 00
	0040	1e 00 0a 03 03 03 80 00

(13) RTB GE0/0/0 发送的LSA类型是LSU报文，包含的LSA类型有Type1 Router-LSA和Type 2 Network-LSA:

32 45.594000	172.16.101.1	224.0.0.5	OSPF	78 LS Acknowledge
33 43.922000	172.16.101.2	224.0.0.5	OSPF	118 LS Update
34 44.594000	172.16.101.1	224.0.0.5	OSPF	98 LS Acknowledge
35 44.922000	172.16.101.2	224.0.0.5	OSPF	82 Hello Packet
36 45.110000	172.16.101.1	224.0.0.5	OSPF	82 Hello Packet
37 45.110000	172.16.101.2	224.0.0.5	OSPF	130 LS Update
38 45.594000	172.16.101.1	224.0.0.5	OSPF	98 LS Acknowledge

> Frame 37: 130 bytes on wire (1040 bits), 130 bytes captured (1040 bits) on interf	0000	01 00 5e 00 00 05 00 e0	fc c6
> Ethernet II, Src: HuaweiTechno_ce:08:78 (00:e0:fc:ce:08:78), Dst: IPv4mcast_05	0010	00 74 00 24 00 00 01 59	c7 35
> Internet Protocol Version 4, Src: 172.16.101.2, Dst: 224.0.0.5	0020	00 05 02 04 00 60 0a 02	02 01
> Open Shortest Path First	0030	00 00 00 00 00 00 00 00	00 00
> OSPF Header	0040	02 01 0a 02 02 02 0a 02	02 01
> LS Update Packet	0050	00 24 01 00 00 01 ac 10	65 07
Number of LSAs: 2	0060	00 01 00 01 02 02 ac 10	65 07
> LSA-type 1 (Router-LSA), len 36	0070	00 02 18 ef 00 20 ff ff	ff 00
> LSA-type 2 (Network-LSA), len 32	0080	01 01	

(14) RTB GE0/0/1 发送的LSA类型是LSU报文，包含的LSA类型有Type3 Summary-LSA:

24 38.297000	172.16.102.1	224.0.0.5	OSPF	98 LS Update
25 38.829000	172.16.102.1	224.0.0.5	OSPF	118 LS Acknowled
26 38.954000	172.16.102.2	224.0.0.5	OSPF	98 LS Acknowled
27 40.907000	172.16.102.2	224.0.0.5	OSPF	118 LS Update
28 41.829000	172.16.102.1	224.0.0.5	OSPF	98 LS Acknowled
29 43.891000	172.16.102.2	224.0.0.5	OSPF	82 Hello Packet
30 45.219000	172.16.102.1	224.0.0.5	OSPF	82 Hello Packet

> Frame 27: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interf	0000	01 00 5e 00 00 05 00 e0	
> Ethernet II, Src: HuaweiTechno_af:23:40 (00:e0:fc:af:23:40), Dst: IPv4mcast_05	0010	00 68 00 1a 00 00	
> Internet Protocol Version 4, Src: 172.16.102.2, Dst: 224.0.0.5	0020	00 05 02 04 00 54	
> Open Shortest Path First	0030	00 00 00 00 00 00 00 00	
> OSPF Header	0040	02 03 c0 a8 c8 64	
> LS Update Packet	0050	00 1c ff ff ff ff	
Number of LSAs: 2	0060	1e 00 0a 03 03 03	
> LSA-type 3 (Summary-LSA (IP network)), len 28	0070	ff 00 00 00 00 02	
> LSA-type 3 (Summary-LSA (IP network)), len 28			

(15) Area 1 首先进行内泛洪，RTA 生成 Type 1 Router-LSA，描述其直连网络，RTB 作为 ABR，收到 Type 1 LSA 后生成 Type 3 Summary-LSA，将 192.168.10.0/24 汇总到 Area 0；然后进行 Area 0 泛洪，传播 Type 3 LSA 到 RTC；最后 Area 2 进行泛洪，RTC 将 Type 3 LSA 泛洪到 Area 2，RTD 学习到该路由。

26	35.844000	172.16.101.2	224.0.0.5	OSPF	130 LS Update
27	36.579000	172.16.101.1	224.0.0.5	OSPF	118 LS Acknowledge
28	36.797000	172.16.101.1	224.0.0.5	OSPF	98 LS Update
29	36.829000	172.16.101.2	172.16.101.1	OSPF	78 LS Acknowledge
30	36.829000	172.16.101.2	224.0.0.5	OSPF	78 LS Acknowledge
31	43.266000	172.16.101.2	224.0.0.5	OSPF	90 LS Update
32	43.594000	172.16.101.1	224.0.0.5	OSPF	78 LS Acknowledge
33	43.922000	172.16.101.2	224.0.0.5	OSPF	118 LS Update
34	44.594000	172.16.101.1	224.0.0.5	OSPF	98 LS Acknowledge

```
> Frame 28: 98 bytes on wire (784 bits), 98 bytes captured (784 bit
> Ethernet II, Src: HuaweiTechno_5e:12:49 (00:e0:fc:5e:12:49), Dst:
> Internet Protocol Version 4, Src: 172.16.101.1, Dst: 224.0.0.5
v Open Shortest Path First
  > OSPF Header
  v LS Update Packet
    Number of LSAs: 1
    > LSA-type 1 (Router-LSA), len 36
```

```
0000 01 00 5e 00 00 05 00 e0 fc 5e
0010 00 54 00 0d 00 00 01 59 c7 6c
0020 00 05 02 04 00 40 0a 01 01 01
0030 00 00 00 00 00 00 00 00 00 00
0040 02 01 0a 01 01 01 0a 01 01 01
0050 00 24 00 00 00 01 ac 10 65 02
0060 00 01
```

22	38.282000	172.16.102.2	172.16.102.1	OSPF	126 LS Update
23	38.282000	172.16.102.2	224.0.0.5	OSPF	130 LS Update
24	38.297000	172.16.102.1	224.0.0.5	OSPF	98 LS Update
25	38.829000	172.16.102.1	224.0.0.5	OSPF	118 LS Acknowled
26	38.954000	172.16.102.2	224.0.0.5	OSPF	98 LS Acknowled
27	40.907000	172.16.102.2	224.0.0.5	OSPF	118 LS Update
28	41.829000	172.16.102.1	224.0.0.5	OSPF	98 LS Acknowled
29	43.891000	172.16.102.2	224.0.0.5	OSPF	82 Hello Packet
30	45.219000	172.16.102.1	224.0.0.5	OSPF	82 Hello Packet
31	54.313000	172.16.102.1	224.0.0.5	OSPF	82 Hello Packet

```
> Frame 27: 118 bytes on wire (944 bits), 118 bytes captured (944
> Ethernet II, Src: HuaweiTechno_af:23:40 (00:e0:fc:af:23:40), Dst:
> Internet Protocol Version 4, Src: 172.16.102.2, Dst: 224.0.0.5
v Open Shortest Path First
  > OSPF Header
  v LS Update Packet
    Number of LSAs: 2
    > LSA-type 3 (Summary-LSA (IP network)), len 28
    > LSA-type 3 (Summary-LSA (IP network)), len 28
```

```
0000 01 00 5e 00 00 05
0010 00 68 00 1a 00 00
0020 00 05 02 04 00 54
0030 00 00 00 00 00 00
0040 02 03 c0 a8 c8 64
0050 00 1c ff ff ff ff
0060 1e 00 0a 03 03 03
0070 ff 00 00 00 00 02
```

(16) Area 2 首先进行内泛洪，RTD 生成 Type 1 Router-LSA ，描述其直连网络，RTC 作为 ABR，收到 Type 1 LSA 后生成 Type 3 Summary-LSA ，将 192.168.30.0/24 和 192.168.200.0/24 汇总到 Area 0；然后进行 Area 0 泛洪，传播 Type 3 LSA 到 RTB；最后 Area 1 进行泛洪，RTB 将 Type 3 LSA 泛洪到 Area 1，RTA 学习到该路由。

23	35.829000	172.16.101.1	172.16.101.2	OSPF	82 LS Request
24	35.829000	172.16.101.1	172.16.101.2	OSPF	66 DB Description
25	35.844000	172.16.101.2	172.16.101.1	OSPF	126 LS Update
26	35.844000	172.16.101.2	224.0.0.5	OSPF	130 LS Update
27	36.579000	172.16.101.1	224.0.0.5	OSPF	118 LS Acknowledge
28	36.797000	172.16.101.1	224.0.0.5	OSPF	98 LS Update
29	36.829000	172.16.101.2	172.16.101.1	OSPF	78 LS Acknowledge
30	36.829000	172.16.101.2	224.0.0.5	OSPF	78 LS Acknowledge
31	43.266000	172.16.101.2	224.0.0.5	OSPF	90 LS Update

```
> Frame 26: 130 bytes on wire (1040 bits), 130 bytes captured (1040
> Ethernet II, Src: HuaweiTechno_ce:08:78 (00:e0:fc:ce:08:78), Dst:
> Internet Protocol Version 4, Src: 172.16.101.2, Dst: 224.0.0.5
v Open Shortest Path First
  > OSPF Header
  v LS Update Packet
    Number of LSAs: 2
    > LSA-type 1 (Router-LSA), len 36
    > LSA-type 2 (Network-LSA), len 32
```

```
0000 01 00 5e 00 00 05 00 e0 fc ce
0010 00 74 00 15 00 00 01 59 c7 44
0020 00 05 02 04 00 60 0a 02 02 02
0030 00 00 00 00 00 00 00 00 00 00
0040 02 01 0a 02 02 02 0a 02 02 02
0050 00 24 01 00 00 01 ac 10 65 02
0060 00 01 00 01 02 02 ac 10 65 02
0070 00 01 1a ee 00 20 ff ff ff 00
0080 01 01
```

26	38.954000	172.16.102.2	224.0.0.5	OSPF	98	LS Acknowled
27	40.907000	172.16.102.2	224.0.0.5	OSPF	118	LS Update
28	41.829000	172.16.102.1	224.0.0.5	OSPF	98	LS Acknowled
29	43.891000	172.16.102.2	224.0.0.5	OSPF	82	Hello Packet
30	45.219000	172.16.102.1	224.0.0.5	OSPF	82	Hello Packet
31	54.313000	172.16.102.1	224.0.0.5	OSPF	82	Hello Packet
32	54.829000	172.16.102.2	224.0.0.5	OSPF	82	Hello Packet
33	63.407000	172.16.102.1	224.0.0.5	OSPF	82	Hello Packet
34	65.766000	172.16.102.2	224.0.0.5	OSPF	82	Hello Packet

>	Frame 28: 98 bytes on wire (784 bits), 98 bytes captured (784 bi	0000	01 00 5e 00 00 05
>	Ethernet II, Src: HuaweiTechno_ce:08:79 (00:e0:fc:ce:08:79), Dst	0010	00 54 00 22 00 00
>	Internet Protocol Version 4, Src: 172.16.102.1, Dst: 224.0.0.5	0020	00 05 02 05 00 40
>	Open Shortest Path First	0030	00 00 00 00 00 00
>	OSPF Header	0040	1e 00 0a 03 03 03
>	LSA-type 3 (Summary-LSA (IP network)), len 28	0050	02 03 c0 a8 c8 64
>	LSA-type 3 (Summary-LSA (IP network)), len 28	0060	00 1c